STATS 598 **10/31/25**

Lecture 11,12

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Topics:

- Maximum likelihood estimation (MLE) for logistic regression in high dimensions [1, 2]
- Random Matrix Theory and Application to Asymptotic of ridge estimator [3]

1 Logistic Regression

Consider the following logistic regression model, $\beta \in \mathbb{R}^p$:

$$x_i \overset{iid}{\sim} \mathcal{N}(0, \Sigma), \quad \mathbb{P}(Y_i = 1 | X_i = x_i) = \frac{\exp(\beta_0 + \beta^\top x_i)}{1 + \exp(\beta_0 + \beta^\top x_i)} := \sigma(\beta_0 + \beta^\top x_i)$$
$$= 1 - \mathbb{P}(Y_i = -1 | X_i = x_i)$$

The log-likelihood and MLE are given by:

$$\ell(b) = \sum_{i=1}^n -\log(1 + \exp(-y_i x_i^\top b))$$
$$\hat{\beta}_{MLE} = \arg\max_{b \in \mathbb{R}^p} \ell(b)$$

1.1 When does MLE exist?

In some cases, MLE does not exist. For example, if the n data points $(\boldsymbol{x}_i, y_i)$ are completely separated in the sense that that there is a linear decision boundary parameterized by $\boldsymbol{b} \in \mathbb{R}^p$ with the property

$$y_i \boldsymbol{x}_i^\top \boldsymbol{b} > 0, \text{ for all } i \in [n] \tag{1}$$

Then the MLE does not exist. To be clear, (1) means that the decision rule that assigns a class label equal to the sign of $\boldsymbol{x}_i' \boldsymbol{b}$ makes no mistake on the sample.

Now we analyze this problem in high dimensions.

In the regime $X \in \mathbb{R}^{n \times p}$ and $\frac{p}{n} \to \kappa$, when $X_i \sim \mathcal{N}(0, \Sigma)$, consider a sequence of problems with β_0 fixed and $\text{Var}(X_i^\top \beta) \to \gamma_0^2$

$$\mathbb{E}[\beta^\top X_i X_i^\top \beta] = \beta^\top \Sigma \beta =: \gamma_0^2$$
$$\Sigma = \frac{1}{n} I_p \Rightarrow \gamma_0^2 = \lim \frac{||\beta||^2}{n}$$

In this case, we can think of γ_0^2 as the singnal strength. This is set so that the log-odds ratio $\beta_0 + \boldsymbol{x}_i^\top \boldsymbol{\beta}$ does not increase with n or p, so that the likelihood is not trivially equal to either 0 or 1 . Instead,

$$\sqrt{\mathbb{E}\left(\beta_0 + \boldsymbol{x}_i^\top \boldsymbol{\beta}\right)^2} \to \sqrt{\beta_0^2 + \gamma_0^2} =: \gamma.$$

Consider the model: Define $(Y, V) \sim F_{\beta_0, X_0}$ such that $(Y, V) \overset{d}{=} (Y, YG)$, here $G \sim \mathcal{N}(0, 1)$, we have:

$$\mathbb{P}[Y = 1|G] = \sigma(\beta_0 + \gamma_0 G) = 1 - \mathbb{P}[Y = 1|G]$$

Theorem 2.1 (Theorem 1 in [1]) *Let $(Y, V) \sim F_{\beta_0, \gamma_0}$ and $Z \sim \mathcal{N}(0, 1)$ be independent random variables*

$$h_{MLE}(\beta_0, \gamma_0) := \min_{t_0, t_1 \in \mathbb{R}} \mathbb{E}[(t_0 Y + t_1 V - Z)_+^2]$$

Here $x_+ := \max(x, 0)$ and $x_+^2 := (x_+)^2$ Then we have:

$$\kappa > h_{MLE}(\beta_0, \gamma_0) \Rightarrow \lim_{n,p \to \infty} \mathbb{P}(\hat{\beta}_{MLE} \textit{ exists}) = 0$$
$$\kappa < h_{MLE}(\beta_0, \gamma_0) \Rightarrow \lim_{n,p \to \infty} \mathbb{P}(\hat{\beta}_{MLE} \textit{ exists}) = 1$$

Note that we can check two extreme cases to validate how the theorem above predict on the existence of MLE. In particular, for $\gamma := \sqrt{\beta_0^2 + \gamma_0^2}$, the case when $\gamma = 0$ (no signal) and $\gamma \to \infty$ (Strong Signal).

Case 1: $\gamma_0 = 0$, $\beta_0 = 0$, since Y is dependent from V and $Y \sim$ Rademacher (Y takes value ± 1 with probability $1/2$), $V \sim \mathcal{N}(0, 1)$, and $t_0 Y + t_1 V - Z$ follows a symmetric distribution.

$$\begin{aligned} h_{MLE}(0, 0) &= \min_{t_0, t_1 \in \mathbb{R}} \mathbb{E}[(t_0 Y + t_1 V - Z)_+^2] \\ &= \min_{t_0, t_1 \in \mathbb{R}} \frac{1}{2} \mathbb{E}[(t_0 Y + t_1 V - Z)_+^2] + \frac{1}{2} \cdot 0 \\ &= \min_{t_0, t_1 \in \mathbb{R}} \frac{1}{2}(t_0^2 + t_1^2 + 1) = \frac{1}{2} \end{aligned}$$

The last line of equality is due to the independence of Y, V, Z.

Hence, the theorem predicts:

$$\frac{p}{n} > 1/2 \Rightarrow \lim_{n,p \to \infty} \mathbb{P}(\hat{\beta}_{MLE} \text{ exists}) = 0$$
$$\frac{p}{n} < 1/2 \Rightarrow \lim_{n,p \to \infty} \mathbb{P}(\hat{\beta}_{MLE} \text{ exists}) = 1$$

Hence, this recovers and extends Cover's result: in the limit where $\beta_0^2 + \boldsymbol{\beta}' \boldsymbol{\Sigma} \boldsymbol{\beta} \to 0$ (this includes the case where y_i is symmetric and independent of $\boldsymbol{x}_i$, we obtain that the phase transition for whether MLE exists is at $\kappa = 1/2$.

Case 2: $\gamma_0 \to \infty$, β_0 is fixed, then Y can be perfectly predicted by V, hence $V \xrightarrow{d} |Z'|$, where $Z' \sim \mathcal{N}(0,1)$
Now if we set $t_0 = 0$, we obtain

$$\lim_{t_1 \to -\infty} \mathbb{E}[(t_1|Z'| - Z)_+^2] = 0$$

If $\beta_0 \to \infty$, $t_1 = 0$ we have

$$\lim_{t_0 \to -\infty} \mathbb{E}[(t_0 - Z)_+^2] = 0$$

In both cases the threoem require $p/n \to 0$ if we wish to guarantee the existence of MLE.

1.2 If MLE exists, what can we say about its asymptotics?

Recall the MLE is the mimimizer of the negative log-likelihood:

$$\ell(b) = \sum_{i \in [n]} (\rho(X_i^\top b) - Y_i X_i^\top b), \quad \rho(t) = \log(1 + \exp(t))$$

Here $X_i \sim \mathcal{N}(0, \frac{I_p}{n})$, $\frac{\|\beta\|^2}{n} \to \gamma^2$. Define the proximal operator as:

$$\text{prox}_{\lambda\rho}(z) = \arg\min_{t \in \mathbb{R}} \left\{ \lambda\rho(t) + \frac{1}{2}(t - z)^2 \right\}.$$

Moreover $(Q_1, Q_2) \sim \mathcal{N}(0, \Sigma)$, where the covariance matrix Σ can be written as:

$$\mathbf{\Sigma}(\alpha, \sigma) = \begin{bmatrix} \gamma^2 & -\alpha\gamma^2 \\ -\alpha\gamma^2 & \alpha^2\gamma^2 + \kappa\sigma^2 \end{bmatrix}$$

The asymptotic behaviour of both the MLE and LRT is characterized by the solution to the following system of equations in $(\alpha, \sigma, \lambda)$:

$$\begin{aligned} \sigma^2 &= \frac{1}{\kappa^2} \mathbb{E}\left[2\rho'(Q_1) \left(\lambda\rho'\left(\text{prox}_{\lambda\rho}(Q_2) \right) \right)^2 \right] \\ 0 &= \mathbb{E}\left[\rho'(Q_1) Q_1 \lambda\rho'\left(\text{prox}_{\lambda\rho}(Q_2) \right) \right] \\ 1 - \kappa &= \mathbb{E}\left[\frac{2\rho'(Q_1)}{1 + \lambda\rho''\left(\text{prox}_{\lambda\rho}(Q_2) \right)} \right] \end{aligned}$$

In the case that the signal strength vanishes i.e. $\gamma = 0$, we have the following simplified system of equations:

$$\begin{cases} \sigma^2 &= \frac{1}{\kappa^2} \mathbb{E}\left[\left(\lambda\rho'\left(\text{prox}_{\lambda\rho}(\tau Z) \right) \right)^2 \right] \\ 1 - \kappa &= \mathbb{E}\left[\frac{1}{1 + \lambda\rho''\left(\text{prox}_{\lambda\rho}(\tau Z) \right)} \right] \end{cases} \qquad \tau^2 := \kappa\sigma^2, \quad Z \sim \mathcal{N}(0,1).$$

When $\kappa \to 1 \Rightarrow \alpha^* \to 1$, here α^* denotes the bias. (Unbiased if $p/n \to 0$)

Theorem 2.2 (Theorem 2 in [2]) *Assume the dimensionality and signal strength parameters κ and γ are such that $h_{MLE}(\gamma) < \kappa$. Assume the logistic model described above where the empirical distribution of $\{\beta_j\}$ converges weakly to a distribution Π with finite second moment. Suppose further that the second moment converges in the sense that as $n \to \infty, \mathrm{Ave}_j\left(\beta_j^2\right) \to \mathbb{E}\beta^2, \beta \sim \Pi$. Then for any pseudo-Lipschitz function ψ of order 2 [1] the marginal distributions of the MLE coordinates obey*

$$\frac{1}{p}\sum_{j=1}^{p} \psi\left(\hat{\beta}_j - \alpha_\star \beta_j, \beta_j\right) \xrightarrow{a.s.} \mathbb{E}\left[\psi\left(\sigma_\star Z, \beta\right)\right], \quad Z \sim \mathcal{N}(0,1),$$

where $\beta \sim \Pi$, independent of Z.

We can hence obtain the asymptotic distribution of each coordinate of the MLE.

Corollary 2.3 (Theorem 3 in [2]) *Let j be any variable such that $\beta_j = 0$. Then in the setting of above theorem, the MLE obeys*

$$\hat{\beta}_j \xrightarrow{\mathrm{d}} \mathcal{N}\left(0, \sigma_\star^2\right).$$

For any finite subset of null variables $\{i_1, \ldots, i_k\}$, the components of $\left(\hat{\beta}_{i_1}, \ldots, \hat{\beta}_{i_k}\right)$ are asymptotically independent.

Finally, we can also obtain the asymptotic distribution of the likelihood ratio statistics.

Corollary 2.4 (Theorem 4 in [2]) *Consider the $LLR \Lambda_j = \min_{\boldsymbol{b}: b_j = 0} \ell(\boldsymbol{b}) - \min_{\boldsymbol{b}} \ell(\boldsymbol{b})$ for testing $\beta_j = 0$. In the setting of Theorem 2, twice the LLR is asymptotically distributed as a multiple of a chi-square under the null,*

$$2\Lambda_j \xrightarrow{\mathrm{d}} \frac{\kappa\sigma_\star^2}{\lambda_\star}\chi_1^2.$$

Also, the LLR for testing $\beta_{i_1} = \beta_{i_2} = \ldots = \beta_{i_k} = 0$ for any finite k converges to the rescaled chi-square $\left(\kappa\sigma_\star^2/\lambda_\star\right)\chi_k^2$ under the null.

We sketch the proof of Theorem 2.2 based on section 4 of [2].

(Sketched). We start on Approximate Message Passing (AMP): Consider a sequence of iterates $\left\{\boldsymbol{S}^t, \hat{\boldsymbol{\beta}}^t\right\}_{t\geq 0}$, with $\boldsymbol{S}^t \in \mathbb{R}^n, \hat{\boldsymbol{\beta}}^t \in \mathbb{R}^p$. Starting from an initial guess β^0, set $S^0 = X\beta^0$. For $t \geq 1$, we inductively define:

$$\hat{\boldsymbol{\beta}}^t = \hat{\boldsymbol{\beta}}^{t-1} + \kappa^{-1}\boldsymbol{X}'\Psi_{t-1}\left(\boldsymbol{y}, \boldsymbol{S}^{t-1}\right)$$
$$\boldsymbol{S}^t = \boldsymbol{X}\hat{\boldsymbol{\beta}}^t - \Psi_{t-1}\left(\boldsymbol{y}, \boldsymbol{S}^{t-1}\right)$$

where the function Ψ_L is applied element-wise and is equal to

$$\Psi_\iota(y, s) = \lambda_\iota r_\iota, \quad r_\iota = y - \rho'\left(\mathrm{prox}_{\lambda_\iota \rho}\left(\lambda_\iota y + s\right)\right)$$

This description requires introducing an augmented sequence $\{\alpha_t, \sigma_t, \lambda_t\}_{t\geq 0}$. With these two extra parameters (α_t, σ_t), we let $\left(Q_1^t, Q_2^t\right)$ be a bivariate normal variable with mean $\mathbf{0}$

[1] A function $\psi : \mathbb{R}^m \to \mathbb{R}$ is said to be pseudo-Lipschitz of order k if there exists a constant $L > 0$ such that for all $\boldsymbol{t}_0, \boldsymbol{t}_1 \in \mathbb{R}^m$, $\|\psi\left(\boldsymbol{t}_0\right) - \psi\left(\boldsymbol{t}_1\right)\| \leq L\left(1 + \|\boldsymbol{t}_0\|^{k-1} + \|\boldsymbol{t}_1\|^{k-1}\right)\|\boldsymbol{t}_0 - \boldsymbol{t}_1\|$.

and covariance matrix $\mathbf{\Sigma}(\alpha_t, \sigma_t)$ defined exactly as in the model. Then starting from an initial pair α_0, σ_0, for $t = 0, 1, \ldots$, we inductively define λ_t

$$\mathbb{E}\left[\frac{2\rho'\left(Q_1^t\right)}{1+\lambda\rho''\left(\text{prox}_{\lambda\rho}\left(Q_2^t\right)\right)}\right] = 1-\kappa$$

Related to Onsager Correction Term in Statistical Physics. Eventially $(\lambda_t, \alpha_t, \sigma_t) \overset{t}{\to} (\lambda_*, \alpha_*, \sigma_*)$ □

2 Extend from analyzing convexity to non-convexity

- CGMT: Require $n, d \to \infty$ then $t \to \infty$
- Approximate Message Passing
- Implicit Bias + Direct Regression

Show that:

$$\theta^{\max} \approx \hat{\beta}_{cvx} \quad \theta^{\max} \to \hat{\beta}_{cvx}$$

Theorem 2.5 *If $y = X\beta, d > n$, then gradient descent with appropriate step size on the least square loss $\frac{1}{2}\|y - X\beta\|$ converges to*

$$\arg\min \|\beta\|_2 \quad s.t.\ y = X\beta$$

As $\lambda \to 0$, we have

$$\arg\min \lambda\|\beta\|_2 + \|y - X\beta\|^2$$

Detour to Random Matrix Theory

Prediction Risk: First we have:

$$\hat{\beta}_\lambda = (X^\top X/n + \lambda I)^{-1} X^\top y/n \quad \hat{\beta}_0 = (X^\top X/n)^+ X^\top y/n$$

The prediction risk is defined as:

$$\mathbb{E}[(X_{new}^\top \hat{\beta}_\lambda - y_{new})^2] = \sigma^2 + \mathbb{E}[(X_{new}^\top \hat{\beta}_\lambda - X_{new}^\top \beta)^2 | X]$$

- If $\frac{d}{n} \to 0$, the second term $\to 0$ almost surely.

$$\mathbb{E}[(X_{new}^\top \hat{\beta}_\lambda - X_{new}^\top \beta)^2 | X] = \lambda^2 \beta^\top (\hat{\Sigma} + \lambda I)^{-1} \Sigma (\hat{\Sigma} + \lambda I)^{-1} \beta + \frac{\sigma^2}{n} \text{tr}[\hat{\Sigma}(\hat{\Sigma} + \lambda I)^{-1} \hat{\Sigma}]$$

By continuous mapping theorem, we have:

$$\text{Bias} \overset{a.s}{\to} \lambda^2 \beta^\top (\Sigma + \lambda I)^{-1} \Sigma (\Sigma + \lambda I)^{-1} \beta = \lambda^2 \|\beta\|^2 \frac{1}{(1+\lambda)^2}$$

$$\text{Variance} \overset{a.s}{\to} \frac{\sigma^2}{n} \text{tr}[\Sigma(\Sigma + \lambda I)^{-2} \Sigma] = \frac{\sigma^2}{n} \frac{d}{1+\lambda}$$

Take $\Sigma = I$.

- If $\frac{d}{n} \to 0$, exist sequence of λ_n s.t. Bias + Variance $\to 0$.

- If $\frac{d}{n} \to \gamma > 0$, we need to understand limiting behaviour of $(\hat{\Sigma} + \lambda I)^{-1}$

Definition 2.6 Let ρ be a measure, $\mathbb{C}^+ = \{z \in \mathbb{C}, Im(z) > 0\}$, $F\{f : \mathbb{C}^+ \to \mathbb{C}\}$. The Stieltjes transform of ρ is defined as $S_\rho : \mathcal{M} \to \mathcal{F}$:

$$S\rho(z) = \int_{\mathsf{R}} (x - z)^{-1} d\rho(x), \quad z \in \mathbb{C}^+$$

Proposition 2.7 (Properties of Stieltjes Transform [4]) *Let ρ be a probability measure on $\mathbb{R}$, then:*

1. *$S\rho$ is an analytic function on $\mathbb{C}^+$.*
2. *For $z \in \mathbb{C}^+$, we have $Im(S\rho(z)) > 0$.*
3. *$|S\rho(z)| \leq \frac{1}{|Im(z)|}$.*
4. *For continuity points of F_ρ, cdf of ρ we have:*

$$\rho([a, b]) = \frac{1}{\pi} \lim_{\eta \to 0^+} \int_a^b Im(S_\rho(x + i\eta)) dx$$

Proof. Proof of 4:

$$\begin{aligned}
\frac{1}{\pi} \lim_{\eta \to 0^+} \int_a^b \mathrm{Im} \left(\int_{\mathbb{R}} \frac{1}{x - \xi - i\eta} d\rho(x) \right) d\xi &= \frac{1}{\pi} \lim_{\eta \to 0^+} \int_a^b \int_{\mathbb{R}} \mathrm{Im} \left(\frac{1}{x - \xi - i\eta} \right) d\rho(x) d\xi \\
&= \frac{1}{\pi} \lim_{\eta \to 0^+} \int_a^b \int_{\mathbb{R}} \frac{\eta}{(x - \xi)^2 + \eta^2} d\rho(x) d\xi \\
&= \frac{1}{\pi} \lim_{\eta \to 0^+} \int_{\mathbb{R}} \int_a^b \frac{\eta}{(x - \xi)^2 + \eta^2} d\xi d\rho(x) \\
&= \frac{1}{\pi} \lim_{\eta \to 0^+} \int_{\mathbb{R}} \left[\arctan \left(\frac{b - x}{\eta} \right) - \arctan \left(\frac{a - x}{\eta} \right) \right] d\rho(x) \\
&= \int_{\mathbb{R}} I_{[a,b]} d\rho(x) \\
&= \rho([a, b])
\end{aligned}$$

□

Corollary 2.8 *If $\mathbb{P}$ and $\mathbb{Q}$ are probability measures, $S\mathbb{P} = S\mathbb{Q}$ if and only if $\mathbb{P} \stackrel{d}{=} \mathbb{Q}$.*

Proof. $\Leftarrow$ is trivial.

$\Rightarrow$: For any continuity point of $F_{\mathbb{P}}$ and $F_{\mathbb{Q}}$, we use the inversion formula:

$$\begin{aligned}
\mathbb{P}([a, b]) &= \frac{1}{\pi} \lim_{\eta \to 0^+} \int_a^b Im(S_{\mathbb{P}}(x + i\eta)) dx \\
&= \frac{1}{\pi} \lim_{\eta \to 0^+} \int_a^b Im(S_{\mathbb{Q}}(x + i\eta)) dx \\
&= \mathbb{Q}([a, b])
\end{aligned}$$

Hence $\mathbb{P} \stackrel{d}{=} \mathbb{Q}$. □

Proposition 2.9 ([4]) *Let $S \subset C^+$ be a countable subset with a limit point in $\mathbb{C}^+$. Then a sequence of probability measures ρ_n converges vaguely to a probabilitiy measure ρ if and only if $s_{\rho_n}(z) \to s_\rho(z)$ for all $z \in S$.*

Now we are ready to apply to Random Matrix Theory:

Theorem 2.10 *Suppose we have a Hermitian $p \times p$ matrix A generated by some fixed distribution on $\mathbb{R}^p$ with n observations, where $p \asymp n$, but $p/n \to c \in (0,1)$. We let $\hat{F}_n(A)$ denote the empirical distribution function of the eigenvalues $\lambda_1(A) \leq \lambda_2(A) \leq \ldots \leq \lambda_p(A)$ of A (which are all real). In other words,*

$$\hat{F}_n(A)(x) := \frac{1}{p} \sum_{i=1}^{p} \mathrm{I}\left\{\lambda_i(A) \leq x\right\},$$

which counts the number of eigenvalues less than or equal to x. This is a well-defined measure. Furthermore,

$$\begin{aligned} s_{\hat{F}_n}(z) &= \int_{\mathbb{R}} \frac{1}{x-z} d\hat{F}_n(x) \\ &= \frac{1}{p} \sum_{i=1}^{p} \frac{1}{\lambda_i(A) - z} \\ &= \frac{1}{p} \operatorname{Tr}(A - zI)^{-1} \end{aligned}$$

since the eigenvalues of $(A - zI)^{-1}$ are given by the values in the summand. Hence, by Proposition above, it suffices to show that if A_n are a sequence of $n \times n$ random matrices that $\frac{1}{p} \operatorname{Tr}(A - zI)^{-1} \to s_\rho(z)$ almost surely to show convergence of the distribution of the eigenvalues.

Theorem 2.11 *Let X_i $i \in [n]$ be iid with mean 0, variance $\sigma^2 I_p$, $\sigma^2 < \infty$. And that:*

$$E[|X_{ik}|^{k+\epsilon}] < \infty, \quad \text{for some } \epsilon > 0$$

We have $\hat{F}_n$ empirical cdf of Y_n. Also assume that $\frac{p}{n} \to c \in (0, \infty)$, then almost surely $\hat{F}_n \to F$, where F is Marcenko-Pastur distribution with parameter γ and scale σ^2.

Define Marchenko-Pastur distribution as:

$$\begin{aligned} \mu(A) &= \begin{cases} (1 - \frac{1}{c}) \mathbb{1}_{0 \in A} + U_{1/c}(A) & c > 1 \\ U_{1/c}(A) & c \leq 1 \end{cases} \\ dU_\lambda(x) &= \frac{1}{2\pi\sigma^2} \frac{\sqrt{(\gamma_+ - x)(x - \gamma_-)}}{\gamma x} \mathbb{1}_{[\gamma_-, \gamma_+]}(x) \\ \lambda_\pm &= \sigma^2 (1 \pm \sqrt{\lambda})^2 \\ x &\in [\lambda_-, \lambda_+] \end{aligned}$$

Proof. Goal: Show that $S_{\hat{F}}(z) \overset{a.s.}{\to} S_u(z)$ In particular, define s_n as the Stieljes transform for $\hat{F}_n$. First, we show that $s_n - \mathbb{E}s_n \to 0$ a.s. and pointwise. Then we show that $\mathbb{E}s_n \to s_U$ where U is the measure defined above, again pointwise. □

Let's examine the statement of this theorem. First, if $c > 1$, it is clear that there has to be point mass at zero, since the matrix is rank $n < p$. It follows that the fraction of zero eigenvalues should be approximately $1 - \frac{1}{c} \approx \frac{p-n}{p}$, so that there are $p - n$ zeros for each matrix (as expected). For $p < n$ (the more interesting case), we see that the pdf is compactly supported on the interval above, and approximately looks peaked towards the lower half. when $c = 1$, the interval is simply $\left[0, 4\sigma^2\right]$, and the density looks like a "quarter circle."

Lemma 2.12

$$S_u(z) = \frac{z + 1 - c - \sqrt{(z-1-c)^2 - 4c}}{2z}$$

Proof using inverse Stieltjes transform.

Lemma 2.13 *Consider the quadratic equation $zS^2 + (z - 1 + c)S + c = 0$, where $S = s(z)$. Then the unique solution is $s(z) = cs_u(z)$, where $s_u(z)$ is the Stieljes Transform of the Marcenko-Pastur distribution with $c = 1$.*

New goal: Shoa that $S_{\hat{F}}$ satisfies the equation in Lemma above

$$\begin{aligned} S_n(z) &= \frac{1}{n}\mathrm{tr}((\hat{\Sigma} - zI)^{-1}) \\ Y_n^{(i)} &= Y_n - \frac{1}{p}X_i X_i^\top = \frac{1}{p}\sum_{j \neq i} X_j X_j^\top \\ G_n^{(i)} &= (Y_n^{(i)} - zI)^{-1} \end{aligned}$$

Lemma 2.14

$$X_k^\top G_n(z) X_k = \frac{X_k^\top G_n^{(k)} X_k}{1 + \frac{1}{n} X_k^\top G_n^{(k)} X_k}$$

Proof. Sherman-Morrison-Woodbury formula.

First, note that

$$\begin{aligned} X_k^\top \mathbf{G_n^{(k)}}(z) \left(\mathbf{G_n}(z)\right)^{-1} &= X_k^\top \mathbf{G_n^{(k)}}(z) \left(\mathbf{Y_n} - zI_p\right) \\ &= X_k^\top \mathbf{G_n^{(k)}}(z) \left(\mathbf{Y_n^{(k)}} - zI_p + \frac{1}{n}X_k X_k^\top\right) \\ &= X_k^\top \mathbf{G_n^{(k)}}(z) \left(\mathbf{Y_n^{(k)}} - zI_p\right) + X_k^\top \mathbf{G_n^{(k)}}(z) \left(\frac{1}{n}X_k X_k^\top\right) \\ &= X_k^\top + X_k^\top \mathbf{G_n^{(k)}}(z) \left(\frac{1}{n}X_k X_k^\top\right) \\ &= \left[1 + \frac{1}{n}X_k^\top \mathbf{G_n^{(k)}}(z) X_k\right)\right] X_k^\top \end{aligned}$$

Rearranging the resulting equation yields

$$X_k^\top \mathbf{G_n^{(k)}}(z) = \left[1 + \frac{1}{n} X_k^\top \mathbf{G_n^{(k)}}(z) X_k\right)\right] X_k^\top \mathbf{G_n}(z)$$

and multiplying on the left by the vector X_k yields

$$X_k^\top \mathbf{G_n^{(k)}}(z) X_k = \left[1 + \frac{1}{n} X_k^\top \mathbf{G_n^{(k)}}(z) X_k\right)\right] X_k^\top \mathbf{G_n}(z) X_k.$$

Rearranging gives the result. □

Lemma 2.15

$$\frac{1}{p}\mathrm{tr}(G_n(z)) = -\frac{1}{nz}\sum_{k=1}^{n} \frac{1}{1 + \frac{1}{n} X_k^\top G_n^{(k)} X_k} + o_p(1)$$

Proof. We compute

$$\begin{aligned}
1 &= \frac{1}{p}\operatorname{Tr}(I_p) \\
&= \frac{1}{p}\operatorname{Tr}\left(\left(\mathbf{Y_n} - zI\right)\mathbf{G_n}(z)\right) \\
&= \frac{1}{p}\operatorname{Tr}\left(\frac{1}{n}\sum_{k=1}^{n} X_k X_k^\top \mathbf{G_n}(z) - z\mathbf{G^n}(z)\right) \\
&= \frac{1}{p}\operatorname{Tr}\left(\frac{1}{n}\sum_{k=1}^{n} X_k X_k^\top \mathbf{G_n}(z)\right) - \frac{z}{p}\operatorname{Tr}\left(\mathbf{G_n}(z)\right) \\
&= \frac{1}{p}\frac{1}{n}\sum_{k=1}^{n} \operatorname{Tr}\left(X_k X_k^\top \mathbf{G^n}(z)\right) - \frac{z}{p}\operatorname{Tr}\left(\mathbf{G_n}(z)\right) \\
&= \frac{1}{p}\frac{1}{n}\sum_{k=1}^{n} \operatorname{Tr}\left(X_k^\top \mathbf{G_n}(z) X_k\right) - \frac{z}{p}\operatorname{Tr}\left(\mathbf{G_n}(z)\right) \\
&= \frac{1}{p}\frac{1}{n}\sum_{k=1}^{n} \left(X_k^\top \mathbf{G_n}(z) X_k\right) - \frac{z}{p}\operatorname{Tr}\left(\mathbf{G_n}(z)\right) \\
&= \frac{1}{p}\frac{1}{n}\sum_{k=1}^{n} \left(\frac{X_k^\top \mathbf{G_n^{(k)}}(z) X_k}{1 + \frac{1}{n} X_k^\top \mathbf{G_n^{(k)}}(z) X_k}\right) - \frac{z}{p}\operatorname{Tr}\left(\mathbf{G_n}(z)\right)
\end{aligned}$$

Rearrange the equation to obtain

$$\begin{aligned}
\frac{z}{p}\operatorname{Tr}\left(\mathbf{G_n}(z)\right) &= \frac{1}{p}\frac{1}{n}\sum_{k=1}^{n} \left(\frac{X_k^\top \mathbf{G_n^{(k)}}(z) X_k}{1 + \frac{1}{n} X_k^\top \mathbf{G_n^{(k)}}(z) X_k}\right) - 1 \\
&= \frac{1}{p}\frac{1}{n}\sum_{k=1}^{n} \left(\frac{X_k^\top \mathbf{G_n^{(k)}}(z) X_k}{1 + \frac{1}{n} X_k^\top \mathbf{G_n^{(k)}}(z) X_k}\right) - \frac{1}{n}\frac{1}{p}\sum_{k=1}^{n} p \\
&= \frac{1}{p}\frac{1}{n}\sum_{k=1}^{n} \left(\frac{X_k^\top \mathbf{G_n^{(k)}}(z) X_k}{1 + \frac{1}{n} X_k^\top \mathbf{G_n^{(k)}}(z) X_k} - p\right).
\end{aligned}$$

Rearrange another time by multiplying through by p and modifying the p factor, we obtain

$$
\begin{aligned}
z \operatorname{Tr}\left(\mathbf{G}_{\mathbf{n}}(z)\right) &= \frac{p}{n} \sum_{k=1}^{n} \left(\frac{\frac{1}{p} X_k^\top \mathbf{G}_{\mathbf{n}}^{(\mathbf{k})}(z) X_k}{1 + \frac{1}{n} X_k^\top \mathbf{G}_{\mathbf{n}}^{(\mathbf{k})}(z) X_k} - 1 \right) \\
&= -\frac{p}{n} \sum_{k=1}^{n} \left(\frac{1 + \frac{p-n}{np} X_k^\top \mathbf{G}_{\mathbf{n}}^{(\mathbf{k})}(z) X_k}{1 - \frac{1}{n} X_k^\top \mathbf{G}_{\mathbf{n}}^{(\mathbf{k})}(z) X_k} \right) \\
&= -\frac{p}{n} \sum_{k=1}^{n} \left(\frac{1}{1 - \frac{1}{n} X_k^\top \mathbf{G}_{\mathbf{n}}^{(\mathbf{k})}(z) X_k} \right) + \frac{p}{n} \frac{p-n}{np} \sum_{k=1}^{n} \frac{X_k^\top \mathbf{G}_{\mathbf{n}}^{(\mathbf{k})}(z) X_k}{1 - \frac{1}{n} X_k^\top \mathbf{G}_{\mathbf{n}}^{(\mathbf{k})}(z) X_k} \\
&= -\frac{p}{n} \sum_{k=1}^{n} \left(\frac{1}{1 - \frac{1}{n} X_k^\top \mathbf{G}_{\mathbf{n}}^{(\mathbf{k})}(z) X_k} \right) + o(1)
\end{aligned}
$$

where the $o(1)$ term comes from the fact that $p/n \to c$ and $p - n = o(np)$ and $n, p \to \infty$. □

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